Mathematics - Undergraduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Susan reads a book at a rate of 1 page every 3 minutes. If her reading rate remains the same, which method could be used to determine the number of minutes...
   (a) add 18 and 3
   (b) subtract 18 from 3, then multiply by 3
   (c) divide 18 by 3
   (d) add 18 and 3, then divide by 3
   (e) subtract 3 from 18
2. A function f(x) is given by f(0)=3, f(2)=7, f(4)=11, f(6)=9, f(8)=3. Approximate the area under the curve y=f(x) between x=0 and x=8 using Trapezoidal rule...
   (a) 85.0
   (b) 60.0
   (c) 65.0
   (d) 45.0
   (e) 55.0
3. Compute $\{}dbinom{85}{82}$.
   (a) 102560
   (b) 252
   (c) 101170
   (d) 100890
   (e) 88440
4. Which of these vowels does NOT have a vertical axis of symmetry?
   (a) U
   (b) E
   (c) Y
   (d) V
   (e) X
5. In a game two players take turns tossing a fair coin; the winner is the first one to toss a head. The probability that the player who makes the first toss...
   (a) 1/3

(b) 1/2

(c) 3/4

(d) 1/5

(e) 1/4

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to ’suppose F(x,y,z)=0. What is $\{}frac{\{}partial x}{\{}partial y} \{}frac{\{}partial y}{\{}partial z} \{}frac{\{}partial z}{\{}partial x}$?’ is ’1.5’.

2. True or False: The correct answer to ’Toss a coin repeatedly until two consecutive heads appear. Assume that the probability of the coin landing on heads is 3/7. Calculate the...’ is ’9.99999’.

3. True or False: The correct answer to ’All of the following statements are true for all discrete random variables except for which one?’ is ’The sum of all probabilities for all possible outcomes is always 1.’.

4. True or False: The correct answer to ’Find the sum of all positive integers less than 196 and relatively prime to 98.’ is ’8232’.

5. True or False: The correct answer to ’For the regression line, which of the following statements about residuals is true?’ is ’The sum of the residuals is always one.’.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Statement 1 — Every field is also a ring. Statement 2 — Every ring has a multiplicative identity.

2. Coloring the edges of a complete graph with n vertices in 2 colors (red and blue), what is the smallest n that guarantees there is either a triangle in red or a 6-clique in blue?

End of Paper